Syllabus for 46-926: Statistical Machine Learning I

Fall 2020

Description: This is the first in a two-part sequence on statistical machine learning. This first course covers tools and approaches for prediction, including both regression and classification. The focus is on understanding the foundations of the methods methods so that they can be both applied and modified. Topics include foundations of supervised learning, regularized and nonparametric regression, bias-variance tradeoffs, model validation and assessment, classification, and tree-based methods. The second course will expand into advanced topics, including boosting and ensemble methods, support vector machines, mixture models and topic modeling, natural language processing, markov decision processes and reinforcement learning, and neural networks. There may be some movement of topics between the two courses as time and interest permits.

Professor: Max G'Sell (sounds like "gazelle")
Email: `mgsell@cmu.edu`. Please post questions on Piazza instead of email if possible.
Office: Baker Hall 132B (but not on campus currently)
Office hours: TBD.

Calendar: Monday/Wednesday 3:20pm-4:40pm.

Text: The following three texts are good references for the course. Note that the first two are available for free online from the authors, and the third is available for free download from SpringerLink when you are on the CMU network.

Elements of Statistical Learning, by Trevor Hastie, Robert Tibshirani, and Jerome Friedman. Springer. ISBN: 978-0-387-84858-7

Introduction to Statistical Learning, by Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani. Springer. ISBN: 978-1-4614-7138-7

Applied Predictive Modeling by Max Kuhn, Kjell Johnson. Springer. ISBN: 978-1-4614-6848-6

Other references for special topics will be given in class.

TAs: Wanshan Li
Vaibhav Goel

TA office hours will be held by Zoom. Times will be announced in class.

In any office hours, if ten minutes pass without any questions, then the session may end.

Course Grade: You are responsible for checking your scores regularly on Canvas, and reporting any errors or discrepancies to us.

Homework and quiz scores will contribute to your final course grade. Your overall homework average will count for 58%, and the quizzes will count for the other 42%.

Computing: This course will rely primarily on python for computation. We will fall back on R for topics which are not well-developed in current python packages. The lectures and homework will assume the general proficiency at programming, simulation, and python that you have developed in your previous courses, but will assume limited familiarity with R. As in learning any new programming language, some exploration and internet searching on your part is assumed.

I would highly recommend using jupyter notebooks for working with python in this course, including for writing up homework assignments.

Homework: Homework will generally be due at the beginning of Wednesday classes. Because we often discuss homework in detail at the beginning of lecture, late homeworks will generally not be accepted after the start of class without prior arrangement.

The homeworks are a key component of this course, and a large amount of learning will occur during the completion of the exercises. There will be instances where the homework is used to extend material that was taught during the lecture. There will be cases where important results will be presented in the homework. These new results should be considered as being equally important to results presented during lecture.

Further, the difficulty level of the homework exercises is, on average, greater than that of the examples done during lecture. It is not a wise use of our limited lecture time to work through many extended examples, and watching me work an exercise is not a replacement for doing the work yourself.

Email: Please do not email TAs with concerns over grading, or any other issues with the course. Instead, please make a private post on Piazza, or email the instructor if you must.

I assume that you check, at least daily, the email account which is linked up to Canvas (in other words, your `@andrew.cmu.edu` address). Email can be used for important announcements, so make sure that you are regularly reading mail sent this account.

Please post any questions regarding homework or the lecture to Piazza. We will do our best to be helpful and respond promptly. *However,* you should be reasonable about expectations of responses to questions on the due dates of homeworks. The time leading up a lecture tends to be the busiest time of the week for me.

Weekly Quizzes: Weekly quizzes will be given on Fridays. The actual quizzes will be time-limited and quite short, but they will be available for 24 hours to accommodate students in different time zones. Your work on the quizzes must be your own. Collaboration is not permitted.
There will be no final exam this mini. The quizzes will serve as a replacement.

Lectures: I expect you to attend all lectures. Also, I assume you hear everything I say during lecture. In other words, if I have to make an important announcement, and I do so during normal lecture time, I am not obligated to deliver the announcement via any other means. **In particular, I often make important announcements at the beginning of lecture, and you are responsible for being aware of these. These announcements are often on a slide titled "Announcements."**

The lecture notes will be delivered via tablet. These will be posted as a .pdf file on the Canvas site the day following the lecture. If needed, I will clarify, expand, and/or correct the notes prior to posting them.

Prior to lecture, I will post a version of the lecture slides with blanks to be filled in. You can print a copy of this prior to lecture to follow along. **These may not be the entirety of what we will cover during lecture.**

You are encouraged to ask questions at any time during the lectures. If I feel that your question ventures into technical details which are not pertinent to the discussion, I may ask you to either talk after class, come to office hours, or send me an email. You should not interpret this as me discouraging these (or any other) questions.

Integrity: Please read CMU's policy on cheating and plagiarism:
`http://www.cmu.edu/policies/documents/Cheating.html`

You are not allowed to utilize any materials from past versions of this course. Any such use will be considered cheating, and treated as such.

Collaboration is a tricky issue. The support and assistance of your classmates can go a long way towards helping you to understand the material. Ultimately, however, you are responsible for preparing yourself for the final exam, later courses, and your future. I encourage you to collaborate on understanding the material and your homework. However, everything you submit must be your own work. Your final writeup should be done independently and you must write your own code for computational problems. You are responsible for understanding everything that you submit. **Please ask me if you have any confusion.** Instances where students copy the work of another student will be treated as cheating.